

Pair Coulomb Density Matrix on a Sphere

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I. INTRODUCTION

We want to find the Pair Coulomb Density Matrix (PCDM) on a Sphere so to extend the work of Ref. [1–3].

We will denote with a Greek index a coordinate basis component and with a Roman index a particle label. We will adopt Einstein summation convention to tacitly assume a sum over repeated Greek indexes. We distinguish between upper contravariant Greek indexes and lower covariant Greek indexes. Passing between the two is accomplished with a contraction with the metric tensor, as usual.

The metric tensor of the Sphere of radius R is

$$||g_{\alpha\beta}|| = R^2 \begin{pmatrix} 1 & 0 \\ 0 & \sin^2 \theta \end{pmatrix}. \quad (1.1)$$

with $g = \det||g_{\alpha\beta}|| = R^4 \sin^2 \theta$.

It is important to realize that on a curved surface the two body problem is generally not separable into a center of mass and a relative one body problem. Therefore, in full generality, we will study the position representation $\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$ of the density matrix $\rho = \exp(-Ht)$, where H is the Hamiltonian operator for the two bodies. These are initially at $\mathbf{r}_1$ and $\mathbf{r}_2$ and at an imaginary time t later they are at $\mathbf{r}'_1$ and $\mathbf{r}'_2$.

Therefore, the PCDM on the Sphere will satisfy the Bloch equation

$$\frac{\partial \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)}{\partial t} = \left(\frac{\hbar^2}{2m_1 \sqrt{g_1}} \nabla_{1\alpha} \sqrt{g_1} \nabla_1^\alpha + \frac{\hbar^2}{2m_2 \sqrt{g_2}} \nabla_{2\alpha} \sqrt{g_2} \nabla_2^\alpha - \phi(\mathbf{r}_1, \mathbf{r}_2) \right) \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t), \quad (1.2)$$

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; 0) = \delta(\mathbf{r}_1 - \mathbf{r}'_1) \delta(\mathbf{r}_2 - \mathbf{r}'_2) / \sqrt{g_1 g_2}, \quad (1.3)$$

where, for particle $i = 1, 2$, $g_i = R^4 \sin^2 \theta_i$, $\nabla_i = (\nabla_{i1}, \nabla_{i2}) = (\partial/\partial\theta_i, \partial/\partial\varphi_i)$ is the gradient respect to the coordinates $\mathbf{r}_i = (r_i^1, r_i^2) = (\theta_i, \varphi_i)$, δ is a two dimensional Dirac delta function, ρ is the pair density matrix, and ϕ is the Coulomb pair potential.

The Coulomb pair potential for particles moving **on** the surface can be chosen as

$$\phi_{\text{on}}(\mathbf{r}_1, \mathbf{r}_2) = Z_1 Z_2 e^2 / d(\mathbf{r}_1, \mathbf{r}_2), \quad (1.4)$$

where $Z_i e$ is the charge of particle i and for the distance between the two points $\mathbf{r}_1$ and $\mathbf{r}_2$, $d(\mathbf{r}_1, \mathbf{r}_2)$, on the Sphere we can either choose the geodesic distance

$$d_g(\mathbf{r}_1, \mathbf{r}_2) = R \arccos[\sin \theta_1 \sin \theta_2 + \cos \theta_1 \cos \theta_2 \cos(\varphi_1 - \varphi_2)], \quad (1.5)$$

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or, viewing the Sphere as embedded in the three dimensional space, the Euclidean distance

$$d_e(\mathbf{r}_1, \mathbf{r}_2) = 2R \sin(d_g/2R). \quad (1.6)$$

For particles living **in** the surface [4] we may choose

$$\phi_{\text{in}}(\mathbf{r}_1, \mathbf{r}_2) = -Z_1 Z_2 e^2 \ln[d_e(\mathbf{r}_1, \mathbf{r}_2)/\ell], \quad (1.7)$$

with ℓ a constant with the dimensions of a distance. We will call ϕ_{on}^g , ϕ_{on}^e , ϕ_{in} the three cases just illustrated.

The free particles, $\phi \rightarrow 0$, pair density matrix can be determined from the work of Bastianelli and Corradini [5]. If $m_1 = m_2 = m$ it will simply be as the square of their one free particle density matrix. We will call this solution ρ_0 . Then our PCDM can be written as

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) = \rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) e^{-P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)}, \quad (1.8)$$

where P is the *inter-action*.

Aim of this work is to tabulate $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$ for the three Coulomb cases illustrated above, P_{on}^g , P_{on}^e , P_{in} . For small $t \ll 1$, for each of these cases, we will compare these exact solutions with the corresponding *primitive approximation* [6]

$$P_{\text{primitive}} = t\phi. \quad (1.9)$$

Even if in general we will be interested in the full $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$ in practice in a many body simulation we may use the *end point approximation* for the inter-action [6] where one only needs the diagonal components $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t) = P(d(\mathbf{r}_1, \mathbf{r}_2); t)$. Moreover on a sphere the distance between any two points $\mathbf{r}_1$ and $\mathbf{r}_2$ is bounded from above: the geodesic distance must be smaller than πR and the Euclidean distance than $2R$.

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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